

Thermal Fatigue of Layered Composite Plates: A Measurement Campaign

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Abstract

Layered composite plates were cycled between ambient and elevated temperature while the centre deflection of each plate was recorded. Three stacking sequences were compared over forty cycles. Deflection grew with cycle count and settled after ten cycles in every sequence; the symmetric stack settled at the lowest value. Residual stiffness fell by between six and fourteen per cent, with the largest fall in the asymmetric stack. The measured stiffness sits consistently below the clamped-edge prediction, which is attributed to the compliance of the fixture rather than to damage within the laminate.

Keywords: thermal fatigue, laminate, deflection, residual stiffness, fixture compliance

1. Introduction

The plates were held at 160 C for 15 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature. Each specimen measures 120 x 80 mm and is 6.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio. Torque on the clamping bolts was set to 12 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 5 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit. Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report. The analytical model assumes a perfectly clamped edge. The measured stiffness sits 14 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

2. Introduction

No delamination was visible under the optical microscope at 100x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 18: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 3 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

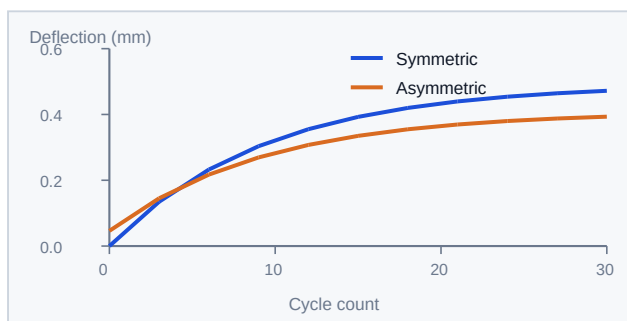


Figure 1: Residual stiffness after thermal cycling, normalised to the as-received value.

Values in this section are means over 5 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

2.1 Motivation

Table 1: measured values, 10 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

The plates were held at 180 C for 25 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Torque on the clamping bolts was set to 12 $\pm$ 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

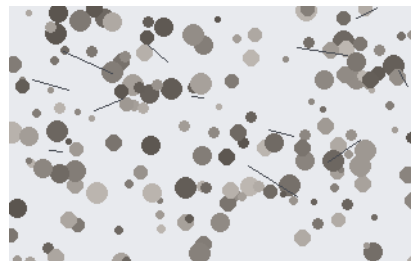


Figure 2: Surface of specimen B-2 after the full campaign.

2.2 Scope of this study

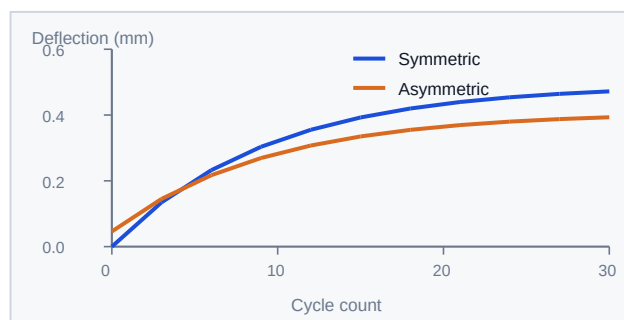


Figure 3: Deflection spread across specimens, by cycle count.

Temperature was logged every 5 s. The uncertainty of the thermocouple is $\pm$ 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

Table 2: measured values, 20 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 11 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

2.3 Related measurements

No delamination was visible under the optical microscope at 100x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 18: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 7 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

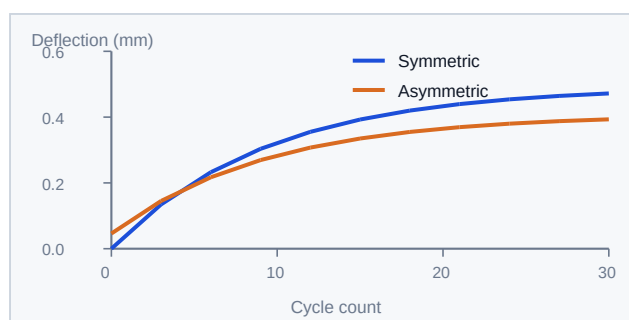


Figure 4: Surface of specimen B-2 after the full campaign.

Values in this section are means over 5 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

3. Materials

Table 3: measured values, 30 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

The plates were held at 160 C for 15 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Each specimen measures 120 x 80 mm and is 4.5 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 14 $\pm$ 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 2 s. The uncertainty of the thermocouple is $\pm$ 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

3.1 Laminate stacking

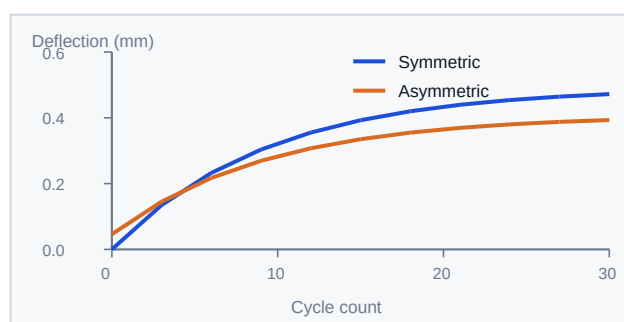


Figure 5: Mean deflection against cycle count for the three stacking sequences.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 6 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

Table 4: measured values, 40 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

No delamination was visible under the optical microscope at 100x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

3.2 Resin system

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 3 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 4 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

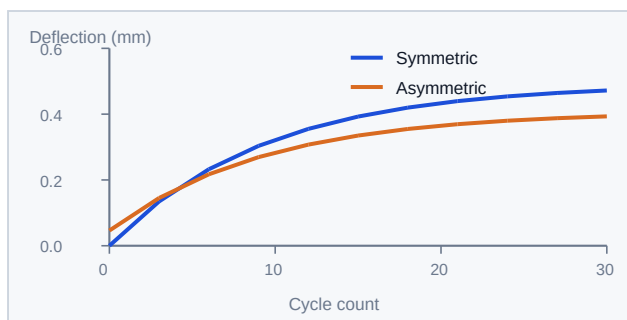


Figure 6: Residual stiffness after thermal cycling, normalised to the as-received value.

The plates were held at 170 C for 15 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

3.3 Specimen preparation

Table 5: measured values, 50 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

Each specimen measures 120 x 80 mm and is 3.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 12 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 1 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

4. Method

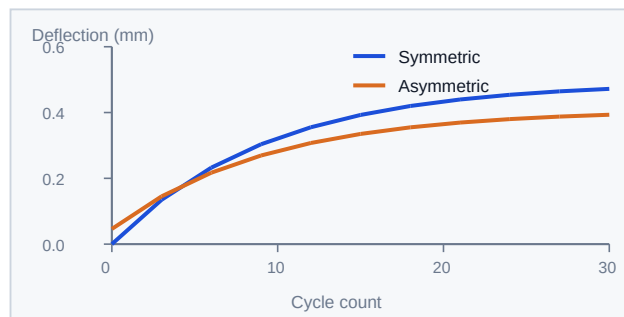


Figure 7: Through-thickness temperature gradient at steady state.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 6 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 50x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Table 6: measured values, 60 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

4.1 Thermal cycling rig

The rig heats the specimen from one side only. A through-thickness gradient of 5 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 6 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

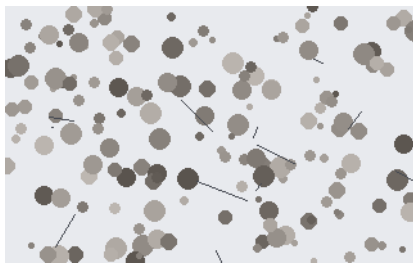


Figure 8: Surface of specimen B-2 after the full campaign.

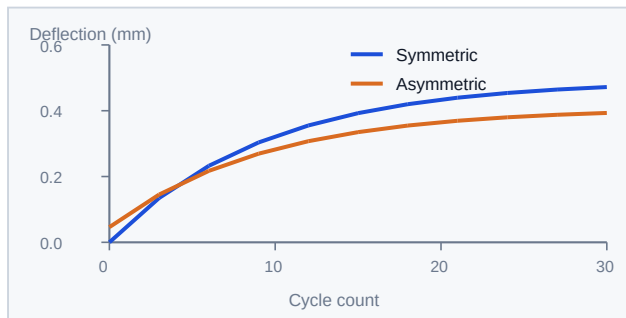


Figure 9: Surface of specimen B-2 after the full campaign.

The plates were held at 180 C for 25 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

4.2 Instrumentation

Table 7: measured values, 70 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

Each specimen measures 120 x 80 mm and is 4.5 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 14 $\pm$ 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 2 s. The uncertainty of the thermocouple is $\pm$ 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

4.3 Uncertainty budget

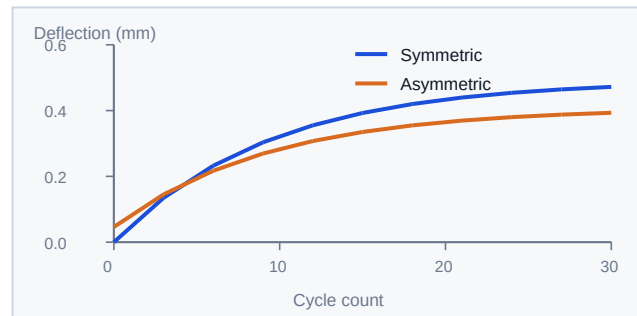


Figure 10: Mean deflection against cycle count for the three stacking sequences.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 11 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 50x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Table 8: measured values, 80 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

5. Results

The rig heats the specimen from one side only. A through-thickness gradient of 5 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 6 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

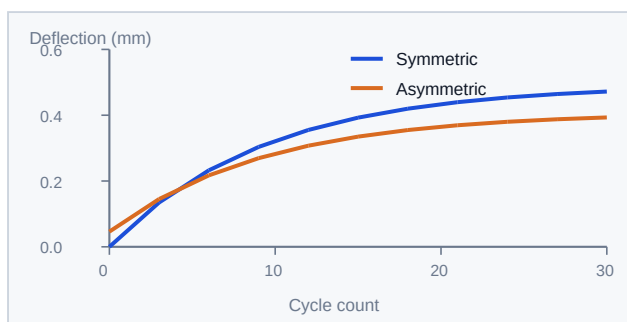


Figure 11: Residual stiffness after thermal cycling, normalised to the as-received value.

Each specimen measures 120 x 80 mm and is 6.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

5.1 Deflection against cycle count

Table 9: measured values, 90 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

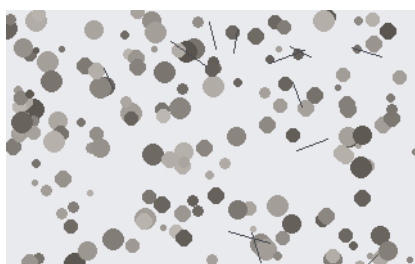


Figure 12: Surface of specimen B-2 after the full campaign.

Torque on the clamping bolts was set to 14 $\pm$ 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 2 s. The uncertainty of the thermocouple is $\pm$ 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

5.2 Effect of stacking order

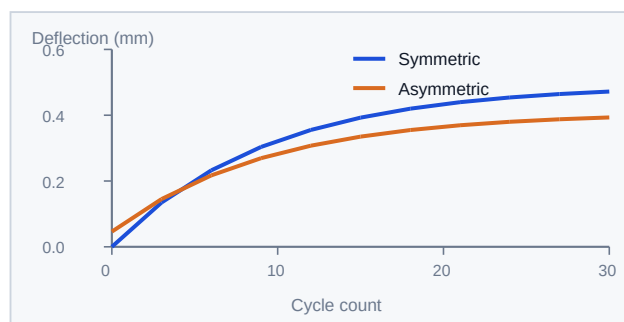


Figure 13: Deflection spread across specimens, by cycle count.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 6 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 50x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Table 10: measured values, 100 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

5.3 Residual stiffness

The rig heats the specimen from one side only. A through-thickness gradient of 5 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 4 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

The plates were held at 170 C for 20 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

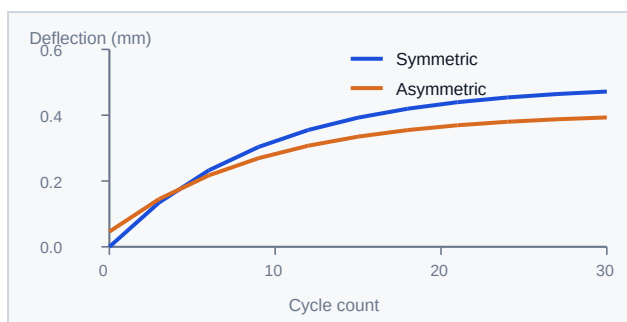


Figure 14: Surface of specimen B-2 after the full campaign.

Each specimen measures 120 x 80 mm and is 3.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

6. Discussion

Table 11: measured values, 110 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

Torque on the clamping bolts was set to 12 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 1 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 6 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

6.1 Comparison with the analytical model

No delamination was visible under the optical microscope at 200x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 18: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

Table 12: measured values, 120 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

The rig heats the specimen from one side only. A through-thickness gradient of 5 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

6.2 Limits of the fixture

Values in this section are means over 4 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

The plates were held at 170 C for 15 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Each specimen measures 120 x 80 mm and is 6.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

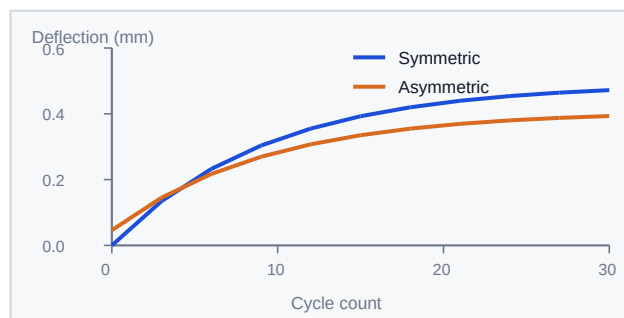


Figure 15: Mean deflection against cycle count for the three stacking sequences.

Torque on the clamping bolts was set to 10 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

7. Conclusion

Table 13: measured values, 130 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

Temperature was logged every 1 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 8 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 50x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

7.1 Findings

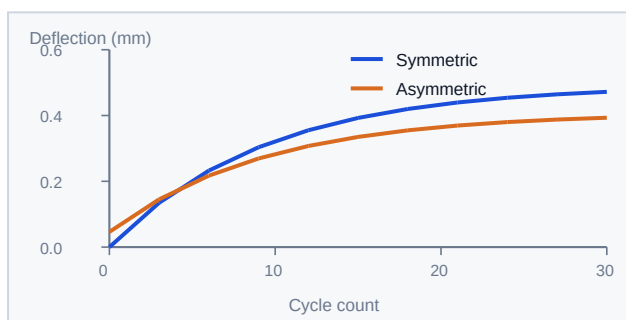


Figure 16: Residual stiffness after thermal cycling, normalised to the as-received value.

Two specimens were removed from the campaign after cycle 18: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 7 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Table 14: measured values, 140 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

Values in this section are means over 5 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

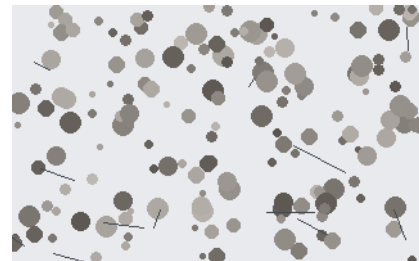


Figure 17: Surface of specimen B-2 after the full campaign.

The plates were held at 180 C for 20 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Each specimen measures 120 x 80 mm and is 4.5 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 10 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 2 s. The uncertainty of the thermocouple is ± 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

7.2 Further work

Table 15: measured values, 150 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 8 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 50x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 12: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

8. Introduction

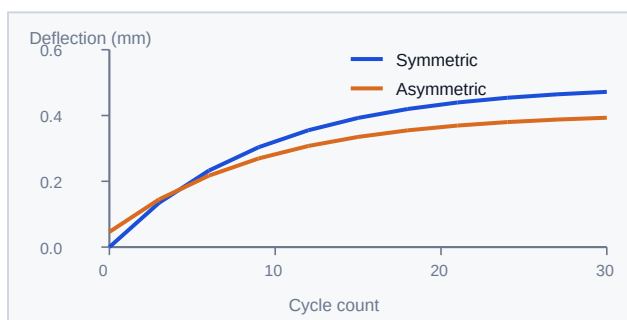


Figure 18: Deflection spread across specimens, by cycle count.

The rig heats the specimen from one side only. A through-thickness gradient of 3 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 6 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

Table 16: measured values, 160 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

The plates were held at 170 C for 15 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

8.1 Motivation

Each specimen measures 120 x 80 mm and is 6.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 10 $\pm$ 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 2 s. The uncertainty of the thermocouple is ± 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

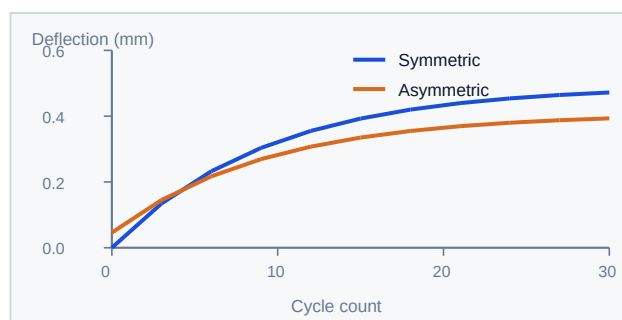


Figure 19: Surface of specimen B-2 after the full campaign.

8.2 Scope of this study

Table 17: measured values, 170 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 8 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 200x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 18: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

8.3 Related measurements

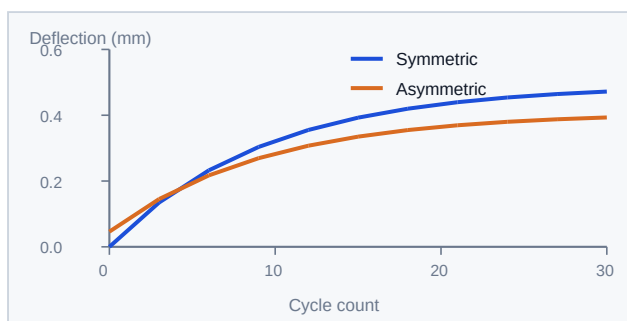


Figure 20: Mean deflection against cycle count for the three stacking sequences.

The rig heats the specimen from one side only. A through-thickness gradient of 5 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Table 18: measured values, 180 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

The plates were held at 190 C for 20 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

9. Materials

Each specimen measures 120 x 80 mm and is 4.5 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 12 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 1 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

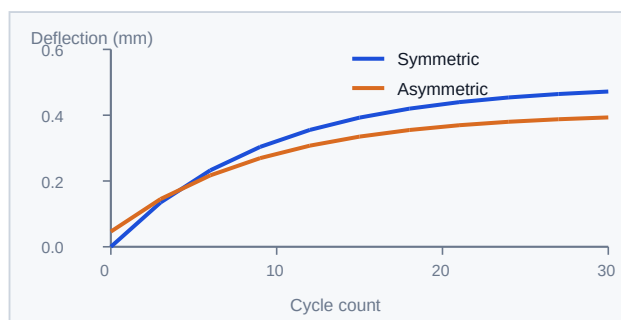


Figure 21: Residual stiffness after thermal cycling, normalised to the as-received value.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

9.1 Laminate stacking

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 14 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 200x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

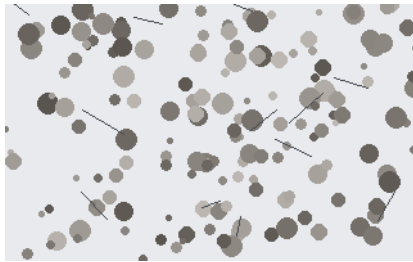


Figure 22: Surface of specimen B-2 after the full campaign.

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

9.2 Resin system

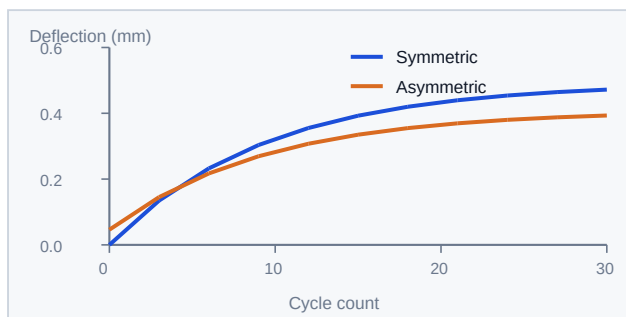


Figure 23: Deflection spread across specimens, by cycle count.

The rig heats the specimen from one side only. A through-thickness gradient of 3 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 5 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

The plates were held at 190 C for 15 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

9.3 Specimen preparation

Each specimen measures 120 x 80 mm and is 4.5 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 14 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 5 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

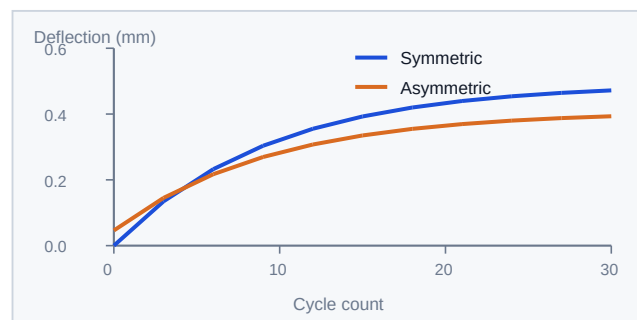


Figure 24: Surface of specimen B-2 after the full campaign.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

10. Method

Table 19: measured values, 190 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 14 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 50x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 7 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

10.1 Thermal cycling rig

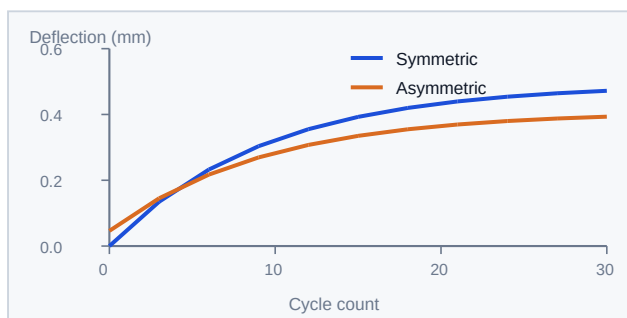


Figure 25: Mean deflection against cycle count for the three stacking sequences.

Values in this section are means over 6 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

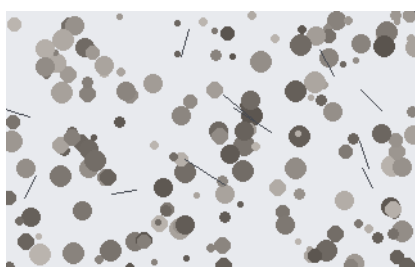


Figure 26: Surface of specimen B-2 after the full campaign.

Table 20: measured values, 200 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

The plates were held at 160 C for 25 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

10.2 Instrumentation

Each specimen measures 120 x 80 mm and is 6.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 12 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 1 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

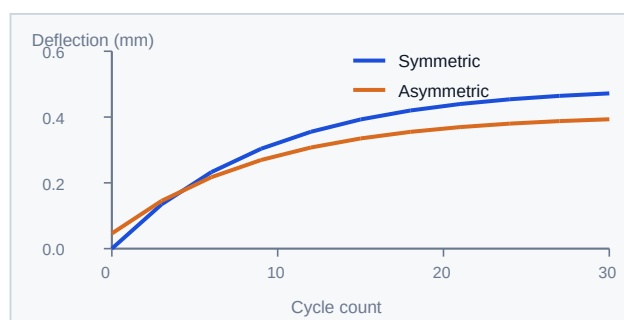


Figure 27: Through-thickness temperature gradient at steady state.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

10.3 Uncertainty budget

Table 21: measured values, 210 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 14 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 100x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 12: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 3 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

11. Results

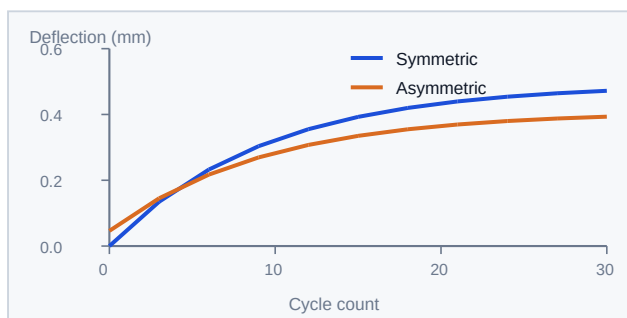


Figure 28: Deflection spread across specimens, by cycle count.

Values in this section are means over 4 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

The plates were held at 170 C for 15 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Table 22: measured values, 220 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

Each specimen measures 120 x 80 mm and is 4.5 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

11.1 Deflection against cycle count

Torque on the clamping bolts was set to 10 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

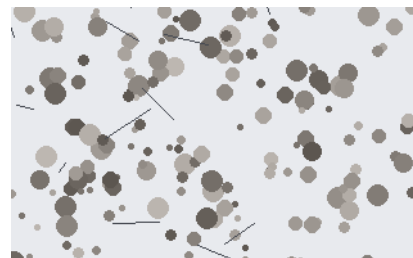


Figure 29: Surface of specimen B-2 after the full campaign.

Temperature was logged every 5 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

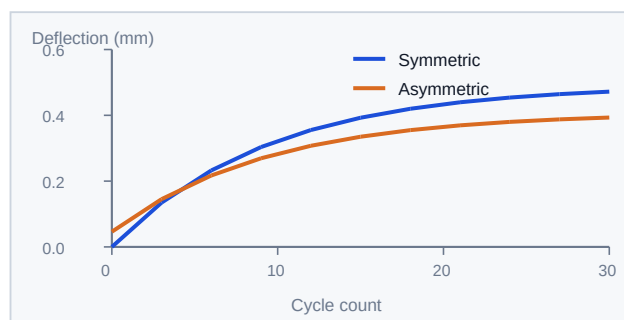


Figure 30: Mean deflection against cycle count for the three stacking sequences.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

11.2 Effect of stacking order

Table 23: measured values, 230 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 11 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 50x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 18: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 5 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

11.3 Residual stiffness

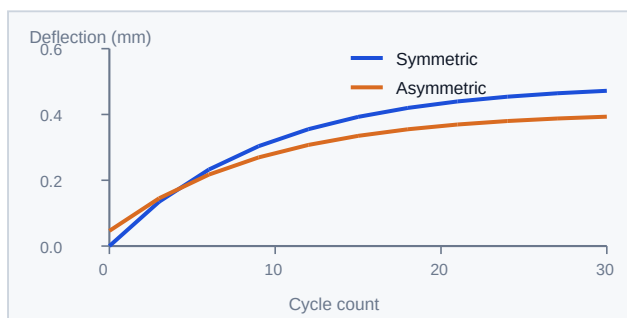


Figure 31: Residual stiffness after thermal cycling, normalised to the as-received value.

Values in this section are means over 5 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

The plates were held at 180 C for 20 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Table 24: measured values, 240 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

Each specimen measures 120 x 80 mm and is 6.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

12. Discussion

Torque on the clamping bolts was set to 10 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 5 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

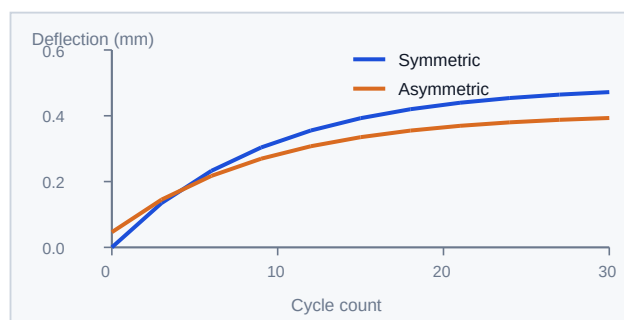


Figure 32: Through-thickness temperature gradient at steady state.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 6 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

12.1 Comparison with the analytical model

Table 25: measured values, 250 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

No delamination was visible under the optical microscope at 100x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 7 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 5 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

12.2 Limits of the fixture

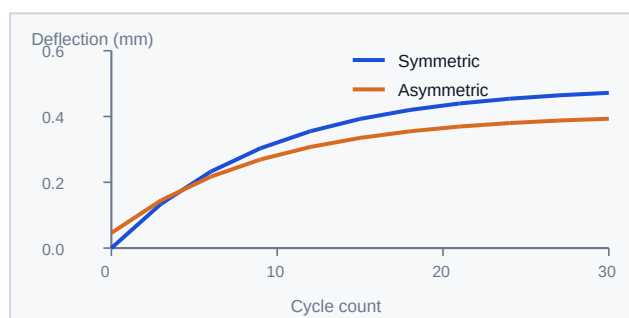


Figure 33: Deflection spread across specimens, by cycle count.

The plates were held at 160 C for 25 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Each specimen measures 120 x 80 mm and is 3.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Table 26: measured values, 260 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

Torque on the clamping bolts was set to 14 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

13. Conclusion

Temperature was logged every 1 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 14 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

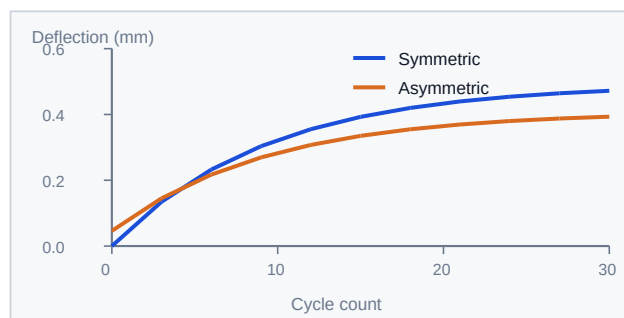


Figure 34: Surface of specimen B-2 after the full campaign.

No delamination was visible under the optical microscope at 50x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

13.1 Findings

Table 27: measured values, 270 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 5 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 5 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

The plates were held at 190 C for 15 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

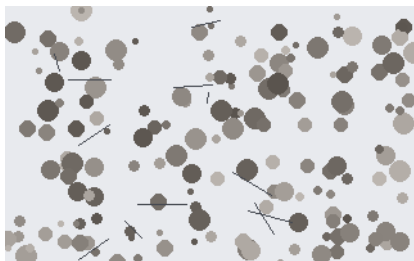


Figure 35: Surface of specimen B-2 after the full campaign.

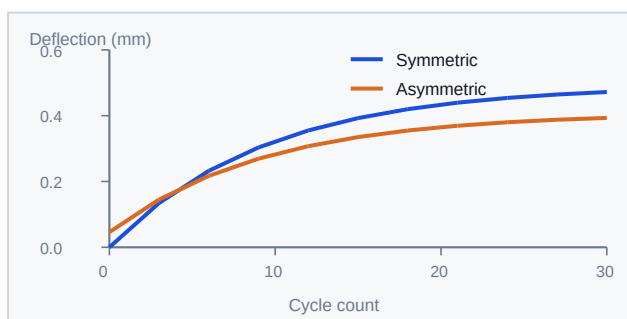


Figure 36: Residual stiffness after thermal cycling, normalised to the as-received value.

Each specimen measures 120 x 80 mm and is 6.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 10 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Table 28: measured values, 280 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

Temperature was logged every 5 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

13.2 Further work

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 6 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 100x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

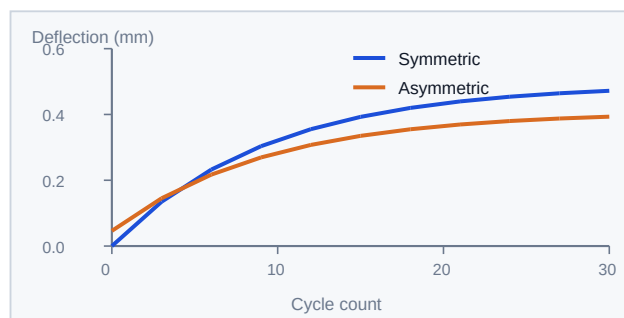


Figure 37: Through-thickness temperature gradient at steady state.

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

14. Introduction

Table 29: measured values, 290 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

The rig heats the specimen from one side only. A through-thickness gradient of 3 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 5 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

The plates were held at 180 C for 15 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Each specimen measures 120 x 80 mm and is 4.5 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

14.1 Motivation

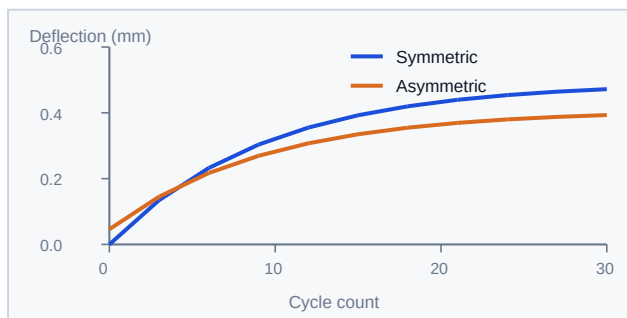


Figure 38: Deflection spread across specimens, by cycle count.

Torque on the clamping bolts was set to 14 +/- 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 1 s. The uncertainty of the thermocouple is +/- 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Table 30: measured values, 300 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

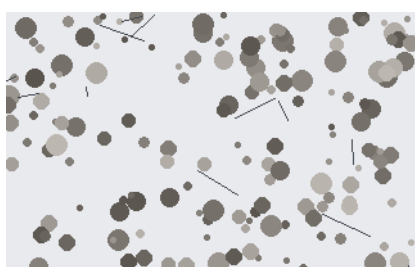


Figure 39: Surface of specimen B-2 after the full campaign.

14.2 Scope of this study

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 8 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 100x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

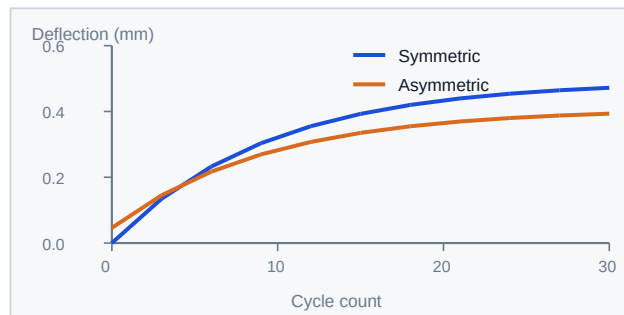


Figure 40: Mean deflection against cycle count for the three stacking sequences.

Two specimens were removed from the campaign after cycle 12: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

14.3 Related measurements

Table 31: measured values, 310 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	±0.02
B-2	20	0.47 mm	±0.03
C-3	30	0.54 mm	±0.04
D-4	40	0.61 mm	±0.05

The rig heats the specimen from one side only. A through-thickness gradient of 5 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

Values in this section are means over 4 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

The plates were held at 180 C for 20 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Each specimen measures 120 x 80 mm and is 3.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

15. Materials

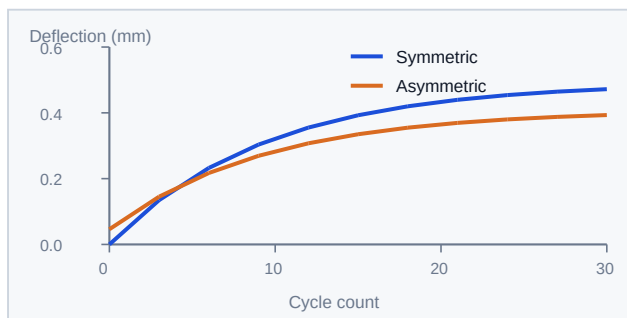


Figure 41: Residual stiffness after thermal cycling, normalised to the as-received value.

Torque on the clamping bolts was set to 10 $\pm$ 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 2 s. The uncertainty of the thermocouple is $\pm$ 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Table 32: measured values, 320 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	$\pm$ 0.02
B-2	20	0.47 mm	$\pm$ 0.03
C-3	30	0.54 mm	$\pm$ 0.04
D-4	40	0.61 mm	$\pm$ 0.05

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

15.1 Laminate stacking

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 14 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

No delamination was visible under the optical microscope at 200x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 12: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

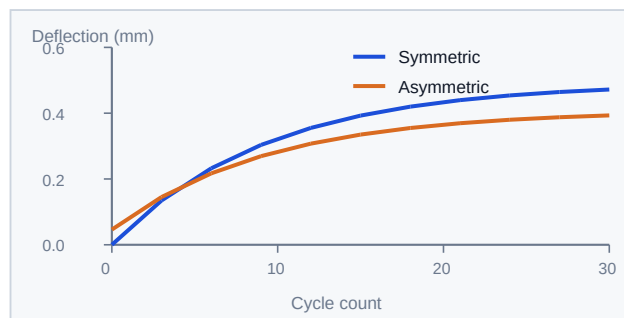


Figure 42: Through-thickness temperature gradient at steady state.

The rig heats the specimen from one side only. A through-thickness gradient of 7 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

15.2 Resin system

Table 33: measured values, 330 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	$\pm$ 0.02
B-2	20	0.47 mm	$\pm$ 0.03
C-3	30	0.54 mm	$\pm$ 0.04
D-4	40	0.61 mm	$\pm$ 0.05

Values in this section are means over 6 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

The plates were held at 180 C for 25 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Each specimen measures 120 x 80 mm and is 3.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 12 $\pm$ 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

15.3 Specimen preparation

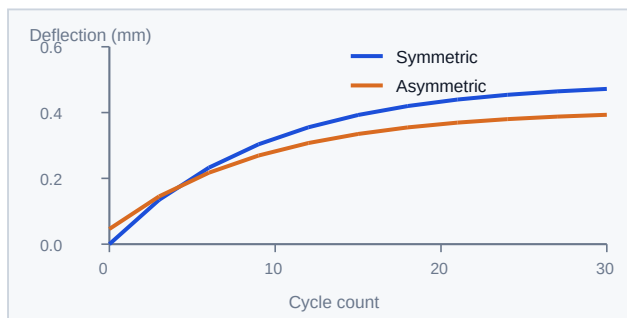


Figure 43: Deflection spread across specimens, by cycle count.

Temperature was logged every 2 s. The uncertainty of the thermocouple is ± 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

Table 34: measured values, 340 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 6 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

16. Method

No delamination was visible under the optical microscope at 50x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 24: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 3 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

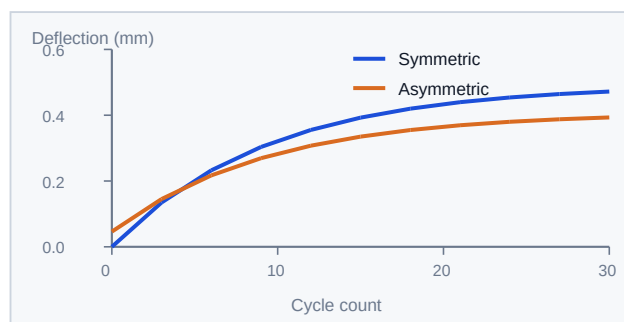


Figure 44: Surface of specimen B-2 after the full campaign.

Values in this section are means over 4 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

16.1 Thermal cycling rig

Table 35: measured values, 350 cycles.

Specimen	Peak temp.	Gradient	Note
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

The plates were held at 170 C for 25 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

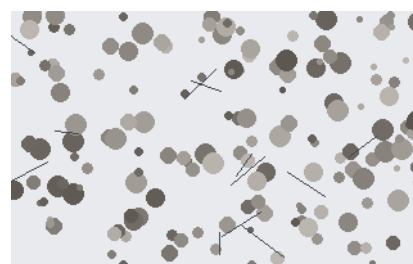


Figure 45: Surface of specimen B-2 after the full campaign.

Each specimen measures 120 x 80 mm and is 6.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 12 $\pm$ 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

16.2 Instrumentation

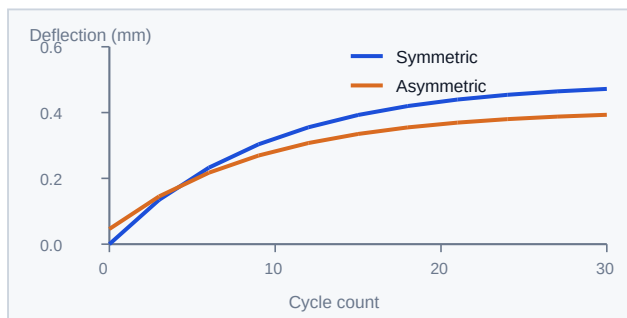


Figure 46: Residual stiffness after thermal cycling, normalised to the as-received value.

Temperature was logged every 2 s. The uncertainty of the thermocouple is ± 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

Deflection grew with the number of cycles and then settled. The change between the tenth and the twentieth cycle was smaller than the measurement uncertainty, so the plates are treated as stable after ten cycles for the purposes of this report.

Table 36: measured values, 360 cycles.

Plate	Cycles	Deflection	Spread
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

The analytical model assumes a perfectly clamped edge. The measured stiffness sits 14 per cent below that prediction, which is consistent with the compliance of the fixture itself rather than with damage in the laminate.

16.3 Uncertainty budget

No delamination was visible under the optical microscope at 200x after the full campaign. That does not rule out matrix microcracking below the resolution of the instrument, and the residual stiffness measurement is the more sensitive indicator.

Two specimens were removed from the campaign after cycle 12: one because a thermocouple detached, and one because the clamping torque had relaxed below the tolerance band. Their partial series are excluded from the averages reported here.

The rig heats the specimen from one side only. A through-thickness gradient of 5 C was measured at steady state, which is small compared with the excursion but not negligible when the residual stress is estimated from the free-edge condition.

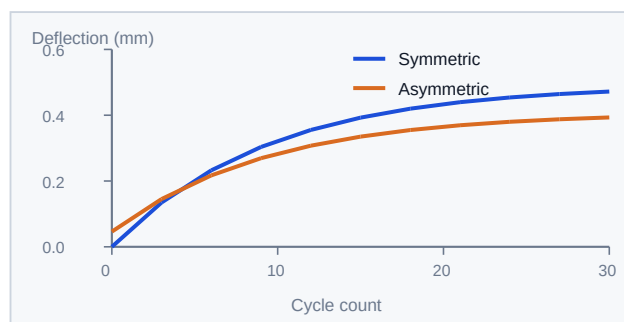


Figure 47: Through-thickness temperature gradient at steady state.

Values in this section are means over 5 specimens unless stated otherwise. The spread is reported as the sample standard deviation rather than the standard error, because the quantity of interest is the behaviour of an individual plate.

17. Results

Table 37: measured values, 370 cycles.

Stacking	Thickness	Stiffness	Change
A-1	10	0.40 mm	± 0.02
B-2	20	0.47 mm	± 0.03
C-3	30	0.54 mm	± 0.04
D-4	40	0.61 mm	± 0.05

The plates were held at 180 C for 20 minutes and cooled in still air before the next cycle began. Deflection was recorded at the centre of each plate with a dial gauge reading to 0.01 mm, and the reading was taken once the surface had returned to ambient temperature.

Each specimen measures 120 x 80 mm and is 3.0 mm thick. The stack is symmetric about the mid-plane, which removes the bending-extension coupling that would otherwise dominate the response at this aspect ratio.

Torque on the clamping bolts was set to 12 $\pm$ 0.5 Nm, which the supplier gives as the limit for this fixture. Above that figure the washers begin to bite into the laminate and the boundary condition is no longer the one the model assumes.

Temperature was logged every 2 s. The uncertainty of the thermocouple is ± 1.5 C over the range used here, and the logger adds a further quantisation error of half a least significant bit.

17.1 Deflection against cycle count